Dongzhi Luo

dongzhluo2@my.cityu.edu.hk | +86 185 8297 2579 | luodongzhi.github.io

Research Interests: Computational Urban Science, Spatial Intelligence, Human Mobility, AI-assisted Urban Planning, Agent-based Modelling

Education

City University of Hong Kong Sep. 2026 – Jul. 2027
Incoming M.Sc. in Physics with Data Modelling and Quantum Technologies — Hong Kong

Lanzhou University Sep. 2024 – Jul. 2025
Graduate Study in Territorial Spatial Planning — **GPA:** 87.6/100 — Lanzhou, China

Northeastern University Sep. 2019 – Jul. 2024
Bachelor's Degree in Urban and Rural Planning — **GPA:** 79.3/100 — Shenyang, China

Publications

Published

- Qiao, W., & **Luo, D.*** (2025). "An autoencoder-based framework for analyzing regional variations in urban green space demand." *Frontiers in Sustainable Cities*, 7, 1642184. DOI

Manuscripts

- **Luo, D.**, Zhang, D., Yang, G., & Wu, C.* "Staged optimization and bounded semantic reasoning for urban planning." Under review at *Environment and Planning B: Urban Analytics and City Science*.
- Wu, C., **Luo, D.**, Mingshu, W.* "Geographic coverage masks uneven lead authorship in coastal sustainability research." Manuscript in preparation for *Nature Sustainability*.

Research Experience

The Hong Kong University of Science and Technology (Guangzhou) Aug. 2025 – Aug. 2026
Research Assistant — Guangzhou, China

- Investigated AI-assisted urban planning, spatial optimization, human mobility, and agent-based modelling; translated planning questions into reproducible Python and GIS workflows.
- Developed spatial-analysis, optimization, and simulation pipelines; contributed to manuscripts, research figures, interactive diagnostics, and collaborative project delivery.

Selected Research Projects

Staged Optimization and Bounded Semantic Reasoning for Urban Planning Aug. 2025 – Jun. 2026

- **Project Lead.** Designed a staged NSGA-II framework separating land-use structure generation from indicator calibration; applied bounded LLM screening to semantic feasibility and exported feasible Pareto solutions for preference experiments.

Research Geography and Lead Authorship in Coastal Sustainability Dec. 2025 – Present

- **Project Lead.** Built a reproducible evidence-mapping workflow comparing study locations with lead-author affiliations; developed a geographic-coverage versus research-leadership framework for a manuscript in preparation.

Dual-Environment Cognitive Urban Agent Simulator 2026 – Present

- **Project Lead.** Modelled GIS-derived actual and cognitive environments with heterogeneous routines, partial observation, and dynamic awareness, familiarity, trust, and uncertainty; implemented interactive SVG graph and map diagnostics.

Older-Adult GPS Map Matching and Activity-Route Extraction 2026 – Present

- **Research Contributor.** Designed a quality-gated HMM/Viterbi workflow for 1,369,928 GPS points from 105 participants; produced person-day activity routes, validated GIS datasets, and an offline interactive web map.

Selected Research Projects (continued)

Functional Enclosure and Public-Space Accessibility Jan. 2026 – Aug. 2026

- **Research Collaborator.** Contributed to a two-city AOI-level index integrating urban morphology, street networks, and aggregate mobility data; supported figure audit and cross-scale reporting for 20,494 residential AOIs.

Urban Green Space Demand Evaluation with Autoencoders Dec. 2024 – Apr. 2025

- **Project Lead.** Integrated Ga2SFCA accessibility, ecological and social indicators, autoencoder feature extraction, and k-means++ clustering; led Python analysis and manuscript development.

Selected Interactive Systems and Game Prototypes

Natural-Language Command RTS Aug. 2026 – Present

- **Product and Systems Designer.** Designed an RTS workflow in which a secretary agent translates natural-language intent into structured, reviewable unit commands; prototyping interfaces and gameplay systems for the 2026 Tencent Game Creation Competition.

Starsea Embers - 2D Pixel Strategy Game 2026

- **Independent Developer.** Built a playable Python prototype with a procedurally generated 24-system galaxy, multi-faction AI, colonies, fleets, wormholes, and a heat-death endgame; validated core logic across 30 random galaxies.

TOEFL Vocabulary Trainer 2026

- **Independent Developer.** Developed and packaged a standalone Windows learning application with a 777-item question bank, per-user progress persistence, and a reproducible build pipeline.

Professional Experience

Liaoning Urban and Rural Construction Planning Design Institute Nov. 2023 – Mar. 2024

Assistant Planner Shenyang, China

- Supported county-level territorial spatial planning through baseline research, land-use indicators, spatial-data organization, technical reports, and regulatory drawing review.

Honors and Awards

3rd Prize, China Graduate Smart City Technology & Creative Design Competition	Sep. 2025
Third-Class Academic Scholarship, Lanzhou University	2024, 2025
3rd Prize, Rural Revitalization Design Competition, Liaoning Civil Architecture Society	Sep. 2023
Outstanding Student Leader, Northeastern University	Dec. 2020

Technical Skills

Programming and Development: Python, JavaScript, HTML/CSS, Three.js, Git; machine learning and AI-assisted rapid prototyping
Geospatial and Quantitative: ArcGIS, QGIS, GeoPandas, MATLAB, SPSS
Agent and LLM Systems: Agent-based modelling, LLM-simulated agents, bounded semantic reasoning
Design and Communication: Figma, Adobe tools, interactive SVG and web mapping
Languages: Chinese (native); English (CET-6: 462)

* Corresponding author.